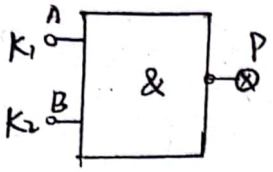
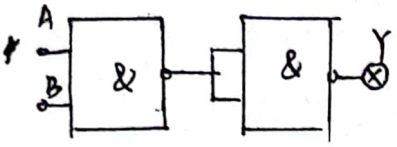
组合数字电路实验原始数据记录

(1) 1)



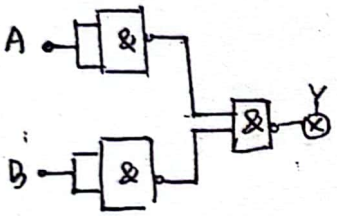
A	B	P
0	0	1
0	1	1
1	0	1
1	1	0

(2) ① 与门



A	B	Y
0	0	0
0	1	0
1	0	0
1	1	1

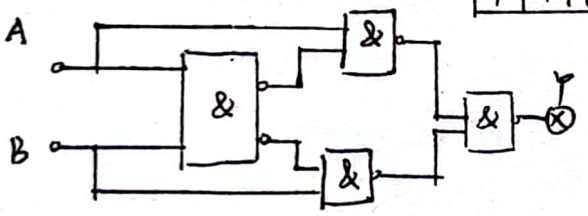
② 或门



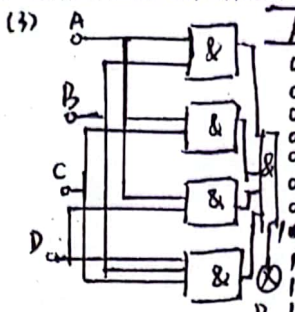
A	B	Y
0	0	0
0	1	1
1	0	1
1	1	1

③ 异或电路

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	0

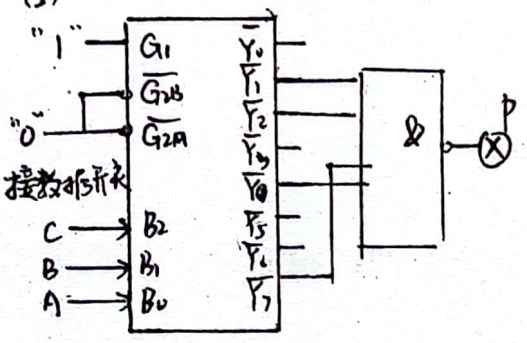


(3)



A	B	C	D	P
0	0	0	0	0
0	0	0	1	0
0	0	1	0	0
0	0	1	1	0
0	1	0	0	0
0	1	0	1	1
0	1	1	0	1
0	1	1	1	1
1	0	0	0	0
1	0	0	1	1
1	0	1	0	1
1	0	1	1	1
1	1	0	0	1
1	1	0	1	1
1	1	1	0	1
1	1	1	1	0

(5)



A	B	C	P
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	0
1	1	0	0
1	1	1	1

(4)

片选端			译码地址			译码输出							
$\overline{G_{2A}}$	$\overline{G_{2B}}$	G_1	B_2	B_1	B_0	$\overline{Y_0}$	$\overline{Y_1}$	$\overline{Y_2}$	$\overline{Y_3}$	$\overline{Y_4}$	$\overline{Y_5}$	$\overline{Y_6}$	$\overline{Y_7}$
H	×	×	×	×	×	H	H	H	H	H	H	H	H
×	H	×	×	×	×	H	H	H	H	H	H	H	H
×	×	L	×	×	×	H	H	H	H	H	H	H	H
L	L	H	L	L	L	L	H	H	H	H	H	H	H
L	L	H	L	L	H	H	L	H	H	H	H	H	H
L	L	H	L	H	L	H	H	L	H	H	H	H	H
L	L	H	L	H	H	H	H	H	L	H	H	H	H
L	L	H	H	L	L	H	H	H	H	L	H	H	H
L	L	H	H	L	H	H	H	H	H	H	L	H	H
L	L	H	H	H	L	H	H	H	H	H	H	L	H
L	L	H	H	H	H	H	H	H	H	H	H	H	L

教师签字

(注:原始数据与实验报告不允许用铅笔记录和书写)

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哈爾濱工業大學

电工电子实验教学中心实验报告

姓名 朱光通 班级 23TS006 学号 2023112380 实验台号 12 成绩 ____

日期 2024.5.30 时间 18:40 教师 侯云鹏 助教 ____

组合数字电路实验

一、实验目的

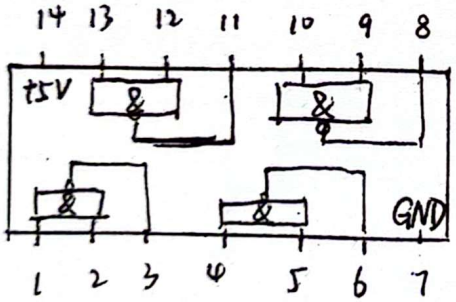
1. 掌握TTL与非门的工作原理和逻辑功能
2. 掌握由与非门实现一些较为复杂的逻辑电路的方法
3. 掌握中规模集成3线-8线译码器74LS138的逻辑功能和使用方法

二、实验设备与平台

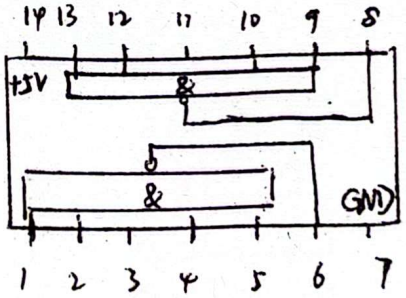
仪器设备：DP832直流稳压电源，Agilent U1241B数字万用表

实验平台：THDZT-1A型电子技术实验装置

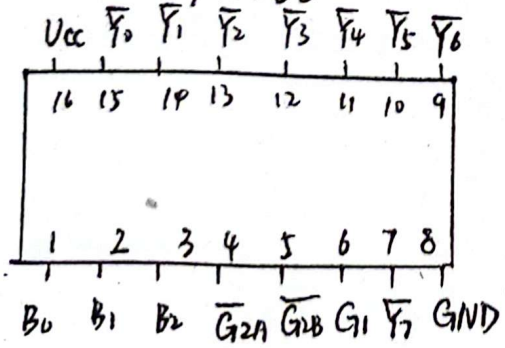
芯片使用：74LS00；74LS20；74LS138



74LS00



74LS20



74LS138

三、实验过程

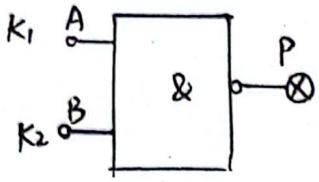
1. 基本任务

(1) 测量与非门功能

1) 输入信号的高低电平由十位逻辑电平输出模块的数据开关给出

2) 利用十位逻辑电平显示模块的电平指示灯观察输出信号的高低电平

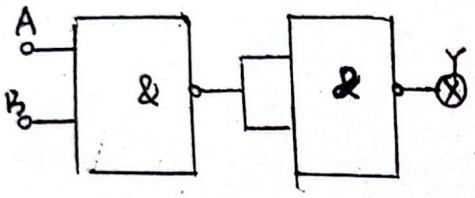
3) 按下图连接电路，改变输入信号状态，观察输出信号.



	A	B	P
1.	0	0	1
2.	0	1	1
3.	1	0	1
4.	1	1	0

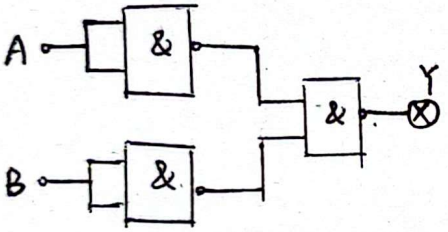
(2) 利用与非门74LS00设计门电路. 绘制原理图并记录真值

① 与门电路



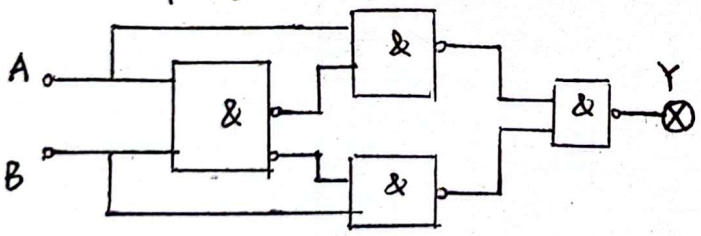
A	B	Y
0	0	0
0	1	0
1	0	0
1	1	1

② 或门电路



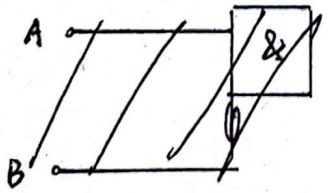
A	B	Y
0	0	0
0	1	1
1	0	1
1	1	1

③ 异或电路



A	B	Y
0	0	0
0	1	1
1	0	1
1	1	0

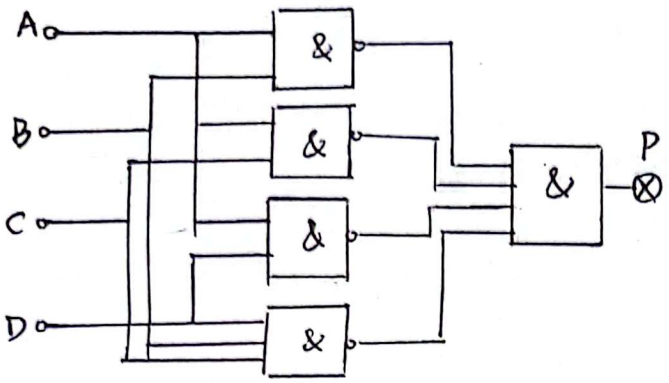
(3) 利用74LS00和74LS20设计4个表决电路.



设计4个表决电路，A用意得2分，其余三个B、C、D同意各得1分，总分大于等于3分时通过，P=1，则电路如图所示

$$P = AB + AC + AD + BCD$$

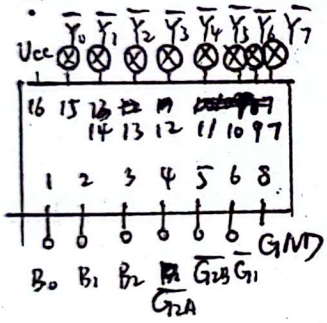
$$= \overline{\overline{AB} \cdot \overline{AC} \cdot \overline{AD} \cdot \overline{BCD}}$$



A	B	C	D	P
0	0	0	1	0
0	0	1	0	0
0	0	1	1	0
0	1	0	0	0
0	1	0	1	0
0	1	1	0	0
0	1	1	1	1
1	0	0	0	0
1	0	0	1	1
1	0	1	0	1
1	0	1	1	1
1	1	0	0	1
1	1	0	1	1
1	1	1	0	1
1	1	1	1	1
0	0	0	0	0

(4) 中规模集成编译器 74LS138功能测试

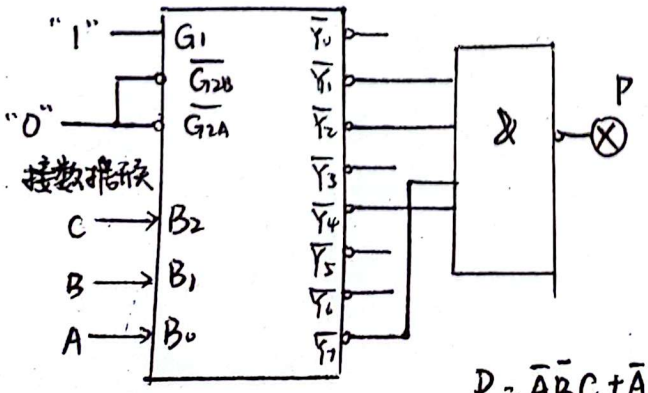
连接如图电路，其功能为：实现3位二进制数向十进制数转换



片选端			译码地址			译码输出							
$\overline{G_{2A}}$	$\overline{G_{2B}}$	G_1	B_2	B_1	B_0	$\overline{Y_0}$	$\overline{Y_1}$	$\overline{Y_2}$	$\overline{Y_3}$	$\overline{Y_4}$	$\overline{Y_5}$	$\overline{Y_6}$	$\overline{Y_7}$
H	×	×	×	×	×	H	H	H	H	H	H	H	H
×	H	×	×	×	×	H	H	H	H	H	H	H	H
×	×	L	×	×	×	H	H	H	H	H	H	H	H
L	L	H	L	L	L	L	H	H	H	H	H	H	H
L	L	H	L	L	H	H	L	H	H	H	H	H	H
L	L	H	L	H	L	H	H	L	H	H	H	H	H
L	L	H	L	H	H	H	H	H	L	H	H	H	H
L	L	H	H	L	L	H	H	H	H	L	H	H	H
L	L	H	H	L	H	H	H	H	H	H	L	H	H
L	L	H	H	H	L	H	H	H	H	H	H	L	H
L	L	H	H	H	H	H	H	H	H	H	H	H	L

(5) 利用 74LS138 设计三开关控制一盏灯电路

连接如图所示电路



A	B	C	P
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	0
1	1	0	0
1	1	1	1

$$P=\bar{A}\bar{B}C+\bar{A}B\bar{C}+A\bar{B}\bar{C}+ABC=\overline{\overline{m_1}\cdot\overline{m_3}\cdot\overline{m_4}\cdot\overline{m_7}}$$

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四、实验结论：

1、掌握了利用74LS00与非门芯片、设计与、或、异或门电路的电路设计、

2、掌握了利用74LS20与74LS00设计4人表决电路。

3、熟练使用74LS138设计复杂电路。

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